CHAPTER 06 / NUTRITION

The nutrition journal

Fuel the work. Understand the energy.

Calories, nutrients and the bigger picture.

How to use this guide

Read the explanations, try the practical examples and complete the worksheet. All training logs and menus are illustrative. Use the website for interactive tools; this document remains useful offline.

01	02	03
Understand	Put into practice	Review

Scope and safety

General education for adults, not a medical diagnosis, personalized treatment or a guaranteed transformation. Stop exercise for sharp pain or unusual symptoms. A qualified clinician, dietitian or coach can help when individual circumstances require assessment.

What you will learn

01. Calories are a unit of energy
02. Your estimate is a starting point
03. Meet the macronutrients
04. Quality, water and timing

01 / Calories are a unit of energy

Food provides energy; your body spends energy at rest, digesting food and moving. Maintenance describes intake that roughly matches expenditure over time. A sustained surplus tends to increase body mass, while a sustained deficit tends to reduce it. Short-term scale changes also reflect water and food in the gut.

PUT IT INTO PRACTICE: Look at trends across weeks, not the morning after one meal.

02 / Your estimate is a starting point

Resting energy expenditure is the energy used at rest. Total daily energy expenditure also includes daily movement, exercise and digestion. The Mifflin–St Jeor equation estimates resting needs; multiplying by an activity factor adds uncertainty. Neither is a laboratory measurement.

PUT IT INTO PRACTICE: Use the calculator, then compare intake and trends before making small adjustments.

03 / Meet the macronutrients

Protein supplies amino acids; carbohydrates support fuel availability; dietary fats support cell function and help absorb fat-soluble vitamins. Carbohydrate and protein provide about four kcal per gram, fat about nine. Fiber supports digestive health and comes from foods such as legumes, whole grains and vegetables.

PUT IT INTO PRACTICE: A meal with rice, dal, curd and vegetables combines several useful nutrients.

04 / Quality, water and timing

Vitamins and minerals support many body processes but do not replace energy or protein. Eat a variety of foods and drink regularly, with more attention in heat or long exercise. Meal timing should fit digestion and your schedule; the whole day matters more than a tiny post-workout window.

PUT IT INTO PRACTICE: Keep a familiar snack available if training falls between meals.

Practical reference

Food composition estimates

Approximate values per 100 g of edible food in the stated condition. Generic estimates vary by brand, recipe and preparation. Price assumptions in the website are editable examples, not live market prices.

Food / condition	kcal	Protein g	Carbs g	Fat g
Rice / cooked	130	2.7	28	0.3
Milk / as sold, whole	61	3.2	4.8	3.3
Curd / plain, whole	61	3.5	4.7	3.3
Egg / boiled, edible portion	155	12.6	1.1	10.6
Chicken breast / cooked, skinless	165	31	0	3.6
Fish / cooked, generic lean fish	128	26	0	2.7
Oats / dry	389	16.9	66.3	6.9
Banana / raw, peeled	89	1.1	22.8	0.3
Peanuts / raw, shelled	567	25.8	16.1	49.2
Almonds / raw	579	21.2	21.6	49.9
Soy chunks / dry, generic label estimate	345	52	33	0.5
Paneer / as sold, generic full-fat estimate	265	18.3	1.2	20.8
Dal / lentils / cooked, plain without oil	116	9	20.1	0.4
Chickpeas / cooked, drained	164	8.9	27.4	2.6
Mixed vegetables / cooked without oil, estimate	65	2.5	13	0.5
Cooking oil / as sold	884	0	0	100

Calculation example

For a hypothetical 25-year-old male weighing 70 kg at 175 cm: resting expenditure = 10 × 70 + 6.25 × 175 − 5 × 25 + 5 = 1,673.75 kcal/day. At an assumed activity multiplier of 1.55, TDEE is approximately 2,594 kcal/day. This is an estimate, not a measurement.

The website uses example goal adjustments of +8% for gradual gain or −10% for gradual loss. Protein is allocated at 1.6 g/kg, fat at 30% of target energy, and carbohydrate receives the remainder. These defaults are teaching assumptions, not individual prescriptions.

A practical student meal

An example plate can combine cooked rice, plain cooked dal, curd and vegetables. Add a protein side that fits preferences and needs. Record the amount and whether it is raw or cooked. Include added oil. A single plate does not establish a complete day of nutrition.

Your study worksheet

Knowledge check

A calorie calculator produces...

A. An exact measurement

B. A starting estimate

C. A guaranteed weight change

Reflect and apply

Prompt	Your response
One idea I can explain in my own words	
One action I can try this week	
What I will observe or measure	
What would make me seek professional guidance	
My next review date	

Chapter summary

Start with an estimate, eat varied foods and adjust from consistent observations.

Answer and explanation

B. Equations and activity factors cannot capture every individual difference.

Sources & limitations

These sources inform the educational principles. Original examples and planning templates are not study findings. No institution listed below endorses this journal. References checked November 2026.

ISSN (2017) — Protein and exercise position stand

https://pmc.ncbi.nlm.nih.gov/articles/PMC5477153/

Mifflin et al. (1990) — Predicting resting energy expenditure

https://pubmed.ncbi.nlm.nih.gov/2305711/

NIDDK — Body Weight Planner and estimation limitations

https://www.niddk.nih.gov/health-information/weight-management/body-weight-planner

USDA — FoodData Central: food composition reference

https://fdc.nal.usda.gov/

U.S. Department of Veterans Affairs — Fiber content of foods and intake guidance

https://www.nutrition.va.gov/docs/UpdatedPatientEd/FiberContentofFoodsAug2023.pdf

Interpretation matters

Research describes groups under particular conditions. Individual responses vary. The website models are simplified illustrations, not precise physiological simulations. Food values and calorie estimates are approximate. Use consistent observation and individualized professional support when needed.